

PERSONAL INFORMATION

Fernando Luz

 Vila Nova de Gaia - Portugal

 prof.fernando.luz@gmail.com

 <https://github.com/fluz>

 <https://www.linkedin.com/in/f-luz/>

POSITION Engineering Manager

WORK EXPERIENCE

Apr/2024 - Present Engineering Lead

AUTODOC

Leading the PDP (Product Detail Page) team through a transition from monolithic architecture to an event-driven microservices ecosystem, owning technical direction and execution for a high-traffic platform.

Driving architectural decisions, delivery practices, and operational standards to enable scalable system evolution, clearer ownership boundaries, and faster, more reliable deployments across several teams.

- Achievements:
- Led the decomposition of PDP into microservices, enabling independent service ownership and contributing to systems handling (25k rps) while establishing a consistent and unified product data layer across domains.
 - Introduced event-driven data pipelines processing around 12M Kafka events/week, enabling downstream teams to consume product data asynchronously and reducing cross-service coupling.
 - Maintained stable latency (100ms) during architectural transition, preventing regressions in user experience while increasing system modularity and deployment independence.
 - Shifted quality ownership to engineering teams and established automated testing as a standard practice, increasing coverage from 25% to 70% and reducing reliance on manual QA across the development lifecycle.

2022 - 2024 Engineering Manager

Talkdesk

Owned delivery and execution across a multi-product portfolio, aligning engineering efforts with customer value and operational reliability.

Led three teams (10+ engineers), evolving team structure, ownership models, and execution practices to improve autonomy, alignment, and delivery throughput.

Drove organizational improvements in quality, cross-team collaboration, and release readiness to enable more predictable and scalable delivery.

- Achievements:
- Drove adoption of automated testing practices across teams, increasing coverage from 50% to 80% and reducing production defects, enabling more predictable releases.
 - Established an end-to-end testing strategy within platform constraints, standardizing release validation and improving stability across multiple product areas.
 - Supported adoption of Agent Flows by strategic customers (e.g., EDP), contributing to product validation and reinforcing the platform's commercial viability.
 - Improved cross-team execution by redefining collaboration models between engineering, product, and QA, reducing delivery friction and enabling more efficient coordination across teams.

2021 - 2022 Senior Software Engineer

Talkdesk

Owned end-to-end delivery of the first production-ready Financial Services platform, contributing to initial market entry and establishing the technical foundation for future expansion.

- Achievements:
- Delivered the first version of Visual IVR for Financial Services, enabling initial customer adoption and validating the product direction.
 - Reduced projected rollout costs by 75% by replacing a third-party frontend with an in-house solution, increasing control over delivery and reducing long-term dependency.
 - Accelerated adoption of engineering practices by delivering internal sessions on Functional Programming and performance testing (K6), increasing team capability and technical alignment.

2018 - 2021 Senior Software Engineer

Capgemini Portugal [ASML Project]

Delivered critical software components for the next-generation ASML lithography platform in a high-reliability environment, balancing feature delivery with testability and long-term maintainability.

Led the Portugal team in expanding unit testing with a **gtest**-based framework, improved a robotics component through a robust **Variant Pattern**, and helped strengthen engineering culture by co-founding an internal Meetup group in Capgemini PT.

- Achievements:
- Delivered the first version of the RYUN (Universal Pick and Place Robot) component with **all planned features** and **100% code coverage**.
 - Accelerated the first phase of the UTTK / ATTEST migration by delivering **400 tests in 10 weeks** against an initial estimate of **24 weeks**.
 - Reduced migration risk in the RYAU component by auditing autotesters and identifying **65% fake tests** (tests without implementation).

2016 - 2018 IT Manager

Technomar Engineering

Led a team of **6 Individual Contributors** responsible for simulator delivery and operational stability for the **Technomar Maritime Simulator**, balancing product commitments with reliability and execution discipline.

Raised engineering maturity by introducing **Scrum**, **Test-Driven Development**, and **Git**, and by coaching the team toward stronger execution and code quality.

Also initiated the certification plan with DNV GL agency and contributed technically to the core hydrodynamic numerical model where that involvement improved simulator fidelity and performance.

- Achievements:
- Improved simulator communication performance by **15%** while also increasing maintainability through changes to the phonon-based communication channel.
 - Delivered a new simulation station end to end, enabling a simulator with 360-degree immersion.
 - Increased delivery transparency and work management discipline by instituting **GitLab** for metrics, bugs, features, backlog, and milestones.

2008 - 2016 Researcher / HPC Software Engineer

Numerical Offshore Tank

Developed and optimized High-Performance Computing applications in the TPN laboratory using C++/C, Python, MPI/sockets, and bash on Linux and Windows.

Combined research, performance engineering, testing practices, and system integration to improve scalability and reliability, and contributed to the creation of **SMH**, a finalist for the 2016 ANP Prize Award for Technological Innovation.

- Achievements:
- Reduced cluster resource usage by **35%** by optimizing parallel execution of the numerical solver based on my Ph.D. research.
 - Reduced execution time by **10%** with **OpenMP** improvements and implementation of **MPI** standards in the parallel version.
 - Improved repository integrity by championing a testing flow and building an initial **CI** setup with CDash and CTest.

2011 - 2016 Graduate Full Professor

Paulista University

Graduate-level teaching and course execution across software engineering and computing topics, combining lectures, labs, assessments, and student mentoring.

Supported hands-on learning through technical projects and clear evaluation criteria, helping students build practical understanding of software development.

EDUCATION AND TRAINING

2010 - 2015 Ph.D. in Computing Engineering

Title Methodology for the execution of parallel applications based on BSP model with heterogeneous tasks

Polytechnic School, University of São Paulo (USP)

2006 - 2010 M.Sc. in Applied Physics

Title Implementation of the MILC package in the study of full QCD

Physics Institute of São Carlos, University of São Paulo (USP)

2001 - 2006 B.Sc. in Physics

Physics Institute of São Carlos, University of São Paulo (USP)

PERSONAL SKILLS

Technical skills

- Agility
- Go
- Jira
- C++
- Bash
- Object-oriented design
- C

Soft skills

- Team player
- Problem Solver
- Ownership
- Time management
- Build solutions
- Lead and deliver complex software systems

ADDITIONAL INFORMATION

VOLUNTEERING

Jun 2022 - Oct 2025 President

Associação de Pais e Encarregados de Educação da Escola Básica Manuel António Pina

Oct 2021 - Jun 2022 Executive Secretary

Associação de Pais e Encarregados de Educação da Escola Básica Manuel António Pina